SALES ANALYSIS

End-to-End Sales Analysis using SQL & Excel

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PostgreSQL | Excel

❑ DATASET OVERVIEW

order_id text	order_date date	customer_id text	customer_name text
ORD0000001	2023-01-31	CUST001504	Vihaan Sharma
ORD0000002	2023-12-30	CUST000178	Pooja Kumar
ORD0000003	2022-05-10	CUST047516	Sneha Singh
ORD0000004	2023-07-18	CUST030059	Vihaan Reddy
ORD0000005	2023-02-04	CUST048677	Aditya Kapoor
ORD0000006	2022-12-31	CUST042705	Karan Sharma
ORD0000007	2024-09-20	CUST037667	Aarav Verma
ORD0000008	2022-11-10	CUST031165	Rohit Kumar
ORD0000009	2024-06-26	CUST026965	Aman Kapoor
ORD0000010	2020-05-01	CUST029472	Aarav Reddy

The dataset was explored and validated using SQL before analysis.

product_id text	product_name text	category text	brand text	quantity integer	unit_price numeric	discount numeric	tax numeric	total_amount numeric	payment_method text	order_status text	city text	state text	country text	seller_id text
P00014	Drone Mini	Books	BrightLux	3	106.59	0	0	319.86	Debit Card	Delivered	Washington	DC	India	SELL019...
P00040	Microphone	Home & Kitchen	UrbanSty...	1	251.37	0.05	19.1	259.64	Amazon Pay	Delivered	Fort Worth	TX	United Stat...	SELL012...
P00044	Power Bank 20000m...	Clothing	UrbanSty...	3	35.03	0.1	7.57	108.06	Debit Card	Delivered	Austin	TX	United Stat...	SELL009...
P00041	Webcam Full HD	Home & Kitchen	Zenith	5	33.58	0.15	11.42	159.66	Cash on Delivery	Delivered	Charlotte	NC	India	SELL011...
P00029	T-Shirt	Clothing	KiddoFun	2	515.64	0.25	38.67	821.36	Credit Card	Cancelled	San Anton...	TX	Canada	SELL014...
P00023	Cookware Set	Books	ReadMore	4	449.73	0	215.87	2017.53	UPI	Delivered	Los Angel...	CA	United Stat...	SELL014...
P00030	Dress Shirt	Clothing	UrbanSty...	2	219.81	0.2	28.14	394.81	UPI	Delivered	Chicago	IL	Australia	SELL016...
P00028	Jeans	Toys & Games	KiddoFun	2	306.51	0.05	29.12	617.73	Debit Card	Pending	Denver	CO	India	SELL005...
P00031	Kids Toy Car	Sports & Outdoo...	Apex	4	146.09	0	46.75	638.14	Debit Card	Delivered	Washington	DC	United Stat...	SELL018...
P00001	Wireless Earbuds	Clothing	Apex	2	278.21	0.1	60.09	565.75	Credit Card	Delivered	Houston	TX	United Stat...	SELL015...

❑ OBJECTIVE

- Analyze overall sales performance.
- Identify top categories and brands.
- Understand customer payment behavior.
- Analyze monthly revenue trends.

❑ PREVIEW

- What is total revenue, orders, and units sold?
- Which category generates the most revenue?
- Which products generate highest revenue?
- Which brands dominate sales?
- How does revenue change over time?
- Which payment method is most used?
- Which category performs best each year?

❑ What is total revenue, orders, and units sold?

```sql
SELECT SUM(total_amount) as total_revenue,
COUNT(*) as total_orders,
SUM(quantity) as units_sold
FROM orders;
```

Data Output | Messages | Notifications

	total_revenue numeric	total_orders bigint	units_sold bigint
1	91825647.92	100000	300140

Total Revenue	Total Orders	Units Sold
9.18 Cr	100K	3L

INSIGHTS:

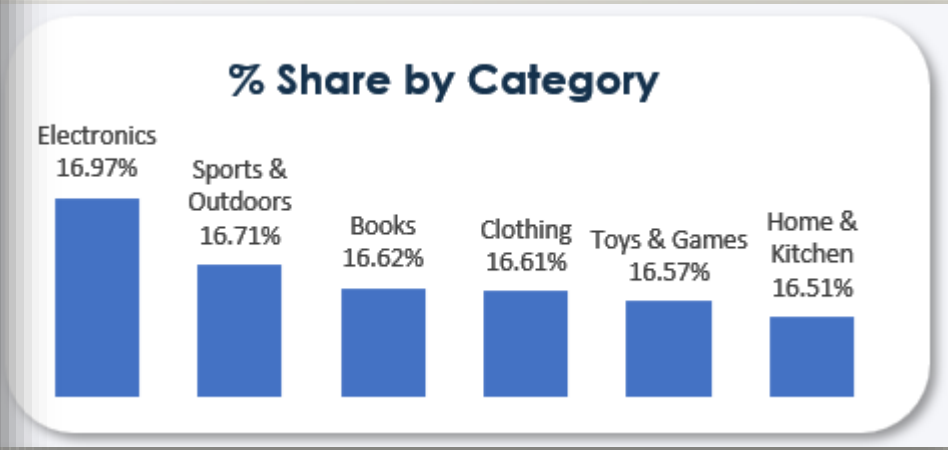
Total revenue is ₹9.18 Cr with 100K orders and 300K units sold, indicating strong sales volume.

❑ Which category generates the most revenue?

```sql
SELECT category, SUM(total_amount) AS revenue,
ROUND(
    SUM(total_amount) * 100.0
    / SUM(SUM(total_amount)) OVER(), 2) AS pct_share
FROM orders GROUP BY category ORDER BY pct_share DESC;
```

Data Output Messages Notifications

	category text	revenue numeric	pct_share numeric
1	Electronics	15584217.18	16.97
2	Sports & Outdoo...	15345571.88	16.71
3	Books	15261837.01	16.62
4	Clothing	15253397.50	16.61



INSIGHTS:

All categories contribute almost equally (~16–17%), showing a balanced sales distribution.

❑ Which products generate highest revenue?

```sql
SELECT product_name,
SUM(total_amount) AS revenue FROM orders
GROUP BY product_name ORDER BY revenue DESC LIMIT 5;
```

Data Output | Messages | Notifications

	product_name text	revenue numeric
1	Memory Card 128GB	1935138.40
2	LED Desk Lamp	1921948.41
3	Mechanical Keyboa...	1906963.54

INSIGHTS:

Top 5 products contribute significantly, with Memory Card and LED Lamp leading sales.

❑ Which brands dominate sales?

```sql
SELECT brand,
SUM(total_amount) AS revenue,
RANK() OVER (ORDER BY SUM(total_amount) DESC) AS rank
FROM orders
GROUP BY brand;
```

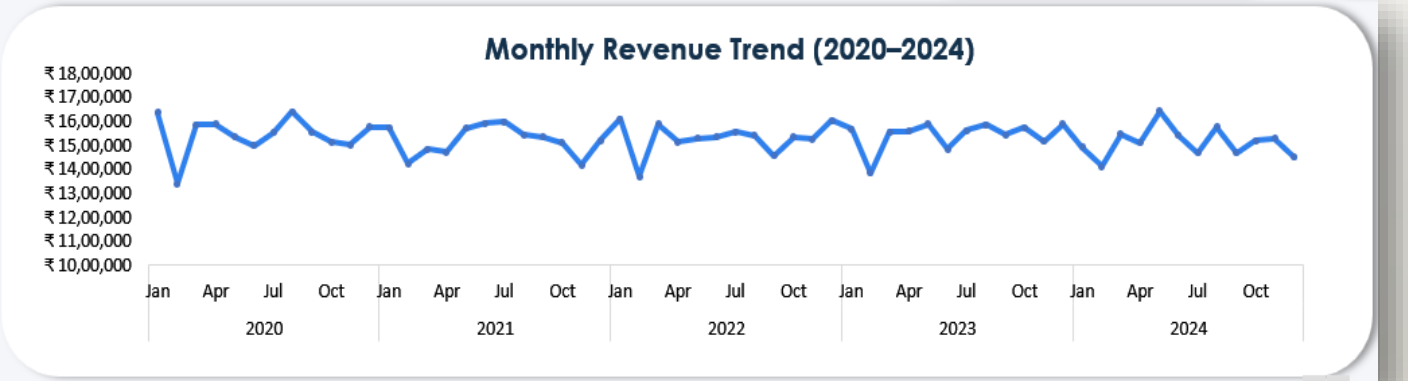
brand text	revenue numeric	rank bigint
CoreTech	9343720.59	1
KiddoFun	9324749.55	2
ReadMore	9278406.63	3
UrbanSty...	9249027.45	4
Zenith	9239960.39	5
Apex	9147604.72	6
NexPro	9078824.97	7
FitLife	9061444.68	8
BrightLux	9056816.11	9
HomeEa...	9045092.83	10

INSIGHTS:

Revenue is evenly distributed across brands, it means no single brand domination.

❑ How does revenue change over time?

```sql
SELECT DATE_TRUNC('month', order_date) AS month,
TO_CHAR(order_date, 'Month') as mon_name,
SUM(total_amount) AS revenue FROM orders
GROUP BY month, mon_name
ORDER BY month;
```



	month timestamp with time zone	mon_name text	revenue numeric
1	2020-01-01 00:00:00+05:30	January	1637069.40
2	2020-02-01 00:00:00+05:30	February	1339388.21
3	2020-03-01 00:00:00+05:30	March	1587125.66
4	2020-04-01 00:00:00+05:30	April	1588347.29
5	2020-05-01 00:00:00+05:30	May	1533834.17
6	2020-06-01 00:00:00+05:30	June	1500269.29
7	2020-07-01 00:00:00+05:30	July	1552981.94
8	2020-08-01 00:00:00+05:30	August	1640874.93
9	2020-09-01 00:00:00+05:30	September	1554682.21
10	2020-10-01 00:00:00+05:30	October	1515403.62
11	2020-11-01 00:00:00+05:30	November	1502232.24
12	2020-12-01 00:00:00+05:30	December	1577655.06
13	2021-01-01 00:00:00+05:30	January	1575253.23
14	2021-02-01 00:00:00+05:30	February	1425365.99
15	2021-03-01 00:00:00+05:30	March	1484165.89
16	2021-04-01 00:00:00+05:30	April	1470885.02
17	2021-05-01 00:00:00+05:30	May	1571842.25
18	2021-06-01 00:00:00+05:30	June	1593398.08
19	2021-07-01 00:00:00+05:30	July	1596668.40
20	2021-08-01 00:00:00+05:30	August	1545324.30
21	2021-09-01 00:00:00+05:30	September	1534912.30
22	2021-10-01 00:00:00+05:30	October	1511118.44
23	2021-11-01 00:00:00+05:30	November	1419480.77

INSIGHTS:

Sales show a stable pattern across months with minor fluctuations, suggesting consistent customer demand throughout the year.

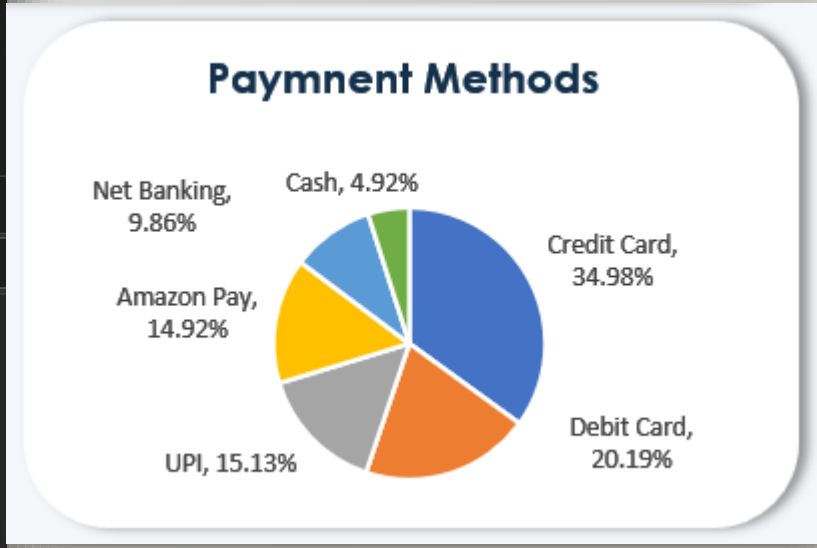
❑ Which payment method is most used?

```sql
SELECT payment_method,
SUM(total_amount) AS revenue,
ROUND(SUM(total_amount)*100.0 / SUM(SUM(total_amount)) OVER(),2) AS pct_share
FROM orders
GROUP BY payment_method
ORDER BY pct_share DESC
```

Data Output Messages Notifications

Showing rows: 1 to 6 Page No: 1

	payment_method text	revenue numeric	pct_share numeric
1	Credit Card	32122158.69	34.98
2	Debit Card	18538678.53	20.19
3	UPI	13896028.55	15.13
4	Amazon Pay	13697498.42	14.92
5	Net Banking	9055674.57	9.86
6	Cash on Delivery	4515609.16	4.92



INSIGHTS:

Credit Card and Debit Card dominate payments, while UPI also has significant share.

❑ Which category performs best each year?

```sql
SELECT year, category, revenue
FROM (
    SELECT EXTRACT(YEAR FROM order_date) AS year,
        category, SUM(total_amount) AS revenue,
        RANK() OVER (
            PARTITION BY EXTRACT(YEAR FROM order_date)
            ORDER BY SUM(total_amount) DESC
        ) AS rank
    FROM orders
    GROUP BY year, category
) t
WHERE rank = 1
ORDER BY year;
```

	year numeric	category text	revenue numeric
1	2020	Electronics	3172686.30
2	2021	Clothing	3177363.85
3	2022	Electronics	3150462.09
4	2023	Electronics	3170585.92
5	2024	Toys & Games	3062914.70

INSIGHTS:

Electronics consistently appears among the top-performing categories across multiple years showing strong demand.

❑ KEY INSIGHTS

- **Total revenue reached ₹9.18 Cr across 100K orders**, indicating strong business performance.
- **Sales** are **evenly distributed** across categories (~16–17%), showing no heavy dependency on a single category.
- ***CoreTech*** and ***KiddoFun*** are the leading brands driving maximum revenue.
- **Credit** and **Debit Cards** are the dominant payment methods, while UPI is also widely used.
- **Monthly revenue trend is stable** with minor fluctuations, indicating consistent demand.
- Top products contribute a significant share of revenue, highlighting product concentration.
- **Electronics** remains a consistently **strong category** across multiple years.